

CFD simulations on high-buoyancy gas dispersion in the wake of an isolated cubic building using steady RANS model and LES

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ARTICLE INFO

Keywords:

Pollutant dispersion
High buoyancy
Building wake
CFD
RANS
LES

ABSTRACT

As the use of emergency generators for buildings increases, the near-field dispersion of high-temperature and high-buoyancy exhaust gas is becoming a concern. We conducted computational fluid dynamics (CFD) simulations using steady Reynolds-averaged Navier-Stokes (RANS) model and large eddy simulation (LES) on the near-field dispersion of high-buoyancy exhaust gas emitted from a building's wake and validated using a wind tunnel experiment. Realizable $k-\epsilon$ 2-layer and wall-adapting local eddy viscosity (WALE) models were used in RANS and LES, respectively. The density difference between the light gas and ambient air in the experiment was reproduced as the temperature difference by an incompressible ideal gas model. RANS model exhibited good agreement with the experimental flow field values for the time-averaged velocity but underestimated the turbulent kinetic energy. LES accurately predicted both the time-averaged velocity and turbulent kinetic energy. For the concentration field, RANS model predicted the region of high time-averaged concentration near the exhaust port but overestimated ground-level values. Meanwhile, LES adequately predicted time-averaged and fluctuating concentrations. Additionally, an investigation of the effect of the turbulent Schmidt number Sc_t in RANS model demonstrated that a small Sc_t increased the overall prediction accuracy, while a large Sc_t compensated for the overestimation on ground-level time-averaged concentrations.

1. Introduction

The prediction of near-field pollution dispersion in urban areas has been a continuous research topic in the past decades. In recent years, with an increase in disaster risks, emergency generators have been widely used owing to their advantage of providing uninterrupted electric power [1,2]. During the operation of these generators, exhaust gas with high temperatures and high concentrations of pollutants are directly released around buildings. The temperatures is over 500°C and the pollutant concentration is over, posing danger to people's health and the surrounding infrastructure [3]. Regular operations of an emergency generator pose a risk in daily life. Moreover, the exhaust ports of emergency generators may not be installed at suitable locations when construction costs and space utilization are restricted. The building downwash effect may trap exhaust gases in a building's wake, impeding the diffusion of the gas and increasing the danger [4]. Therefore, an evaluation of the environmental impact should be conducted and countermeasures devised on exhaust gas dispersed from emergency

generators.

Among the methods of dispersion prediction in the near-field, wind tunnel experiments (WTEs) use real fluids to accurately reproduce physically. WTEs have been considered potent methods from their early stage [5–8] and have recently been used to investigate the dispersion characteristics under various conditions including stack height [9], surrounding building configuration [10,11], thermal stability [12], rooftop structure [13], and wall roughness [14]. However, as Tominaga and Stathopoulos [15] mentioned, WTEs can provide limited information on complicated turbulent diffusion processes. In addition, the buoyancy conditions of buoyant pollutants such as high-temperature exhaust gas reproduced by WTEs are limited by the restriction of similarity criteria [16].

With the development of numerical simulations, computational fluid dynamics (CFD) has been widely used in near-field dispersion prediction [17,18]. A commonly adopted method involves conducting CFD simulations after validating the WTE value. In the CFD technique, direct numerical simulation (DNS) can provide the most detailed spatial information, while a heavy computation burden restricts its use within the

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<https://doi.org/10.1016/j.buildenv.2020.107478>

Received 20 August 2020; Received in revised form 13 November 2020; Accepted 19 November 2020

Available online 26 November 2020

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Nomenclature			
C	dimensionless concentration	Δt	time step
$\bar{C}$	dimensionless time-averaged concentration	T	temperature
C_0	reference concentration	T_a	ambient temperature
C_{raw}	measured concentration	T_s	gas temperature
C_{SD}	dimensionless standard deviation of the concentration	U, V, W	time-averaged velocities
CFD_i	CFD prediction value for sample i	u', v', w'	fluctuation velocities
d_s	diameter of the exhaust port	W_s	exhaust velocity
EXP_i	experiment value for sample i	$\overline{u'v'}$	$\overline{v'w'}$ Reynolds stress
Fr	local Froude number	$\overline{u'_i \phi'}$	turbulent scalar flux
g	acceleration due to gravity	x, y, z	Cartesian coordinates
H	building height	Greek symbols	
h_s	height of exhaust port	α	power-law exponent
I_U	turbulence intensity of streamwise velocity	ε	turbulent dissipation rate
k	turbulent kinetic energy	ν	kinematic viscosity
N	number of data points	ν_t	turbulent kinematic viscosity
Pr_t	turbulent Prandtl number	ρ	density
Re	Reynolds number	ρ_a	ambient air density
Sc_{SGS}	sub-grid scale Schmidt number	ρ_s	gas density
Sc_t	turbulent Schmidt number	φ	mass concentration
t	time		

range of unrevealing the transport mechanism [19]. Meanwhile, large eddy simulation (LES), detached eddy simulation (DES), and Reynolds-averaged Navier-Stokes (RANS) models are utilized more frequently for pollutant dispersion. LES (and DES) can precisely capture the unsteady variation in flow and concentration fields. Many studies using LES were conducted to compare the prediction accuracy with RANS [20–25], examine convective and turbulent mass fluxes [26–28], analyze the effect of thermal stabilities [29,30], and evaluate the effect of computational parameters [31,32]. Nevertheless, the steady RANS approach can provide a reasonable prediction of time-averaged flow and concentration fields at a lower computation cost, and it is widely used to evaluate the effects of building configurations [33–35] and exhaust conditions [9,36].

However, the dispersion characteristics of high-buoyancy gas can be different with neutrally buoyant gas because the buoyancy effect during the dispersion can be larger than the inertia of gas at exhaust port. Meanwhile, only a few studies on the WTEs of high-buoyancy gas dispersion were conducted. Tominaga et al. [37] compared the dispersion of light, neutrally buoyant, and heavy gases. Lin et al. [16] compared high-buoyancy gas with a neutrally buoyant gas at different exhaust ratios, and reported that although high-buoyancy can help dilute pollutant mean concentration, it lead to a stronger concentration fluctuation. Using CFD simulations, Tominaga and Stathopoulos [38] conducted steady RANS computations on the light, neutral, and heavy gas dispersion around an isolated building and compared the results with WTE. They observed that turbulent diffusion was underestimated for light gases and overestimated for heavy gases. Additionally, they reported that a small value of turbulent Schmidt number (Sc_t) can improve the overall prediction performance of light gases. However, the local, particularly near-source, dispersion characteristics affected by Sc_t were not investigated. Olvera et al. [39] observed that plume buoyancy can affect the size and shape of the wake region, thereby affecting the concentration dispersion. Hu et al. [40] conducted an LES on the dispersion of buoyancy-driven smoke soot and carbon monoxide gas in the urban street canyon and investigated the relationship between re-entrainment and wind velocity. Tong and Zhang [41] conducted an LES on exhaust gas dispersion from a diesel backup generator and evaluated the effect of the exhaust momentum ratio and building downwash on the quality of local air. However, to the author's knowledge, no studies have been reported in conducting CFD simulations

using LES on high-buoyancy gas dispersion and validating the accuracy using WTEs.

Therefore, in this study, we conducted LES and steady RANS simulations on the high-buoyancy gas dispersion around an isolated building to clarify CFD performance. The accuracy of LES and RANS computations on the flow and concentration fields were validated using WTEs using light gas. Moreover, concentration fluctuation was validated in the LES. Additionally, we investigated the effect of Sc_t in the RANS model on both the overall prediction performance and local dispersion characteristics.

The remainder of this paper is organized as follows. The experimental setup and validation scenarios are introduced in Section 2. The simulation setup is presented in Section 3. Section 4 presents the results of the flow field and concentration field predicted using RANS model and LES and the validation using WTEs. Subsequently, the effect of Sc_t in the RANS computations is investigated. In Section 5, the overall performance is discussed using validation metrics. Finally, conclusions and perspectives are presented in Section 6.

2. Experimental setup

Lin et al. [16] conducted WTEs on a high-buoyancy gas dispersion around an isolated cubic building using a boundary layer wind tunnel at the Institute of Industrial Science, University of Tokyo. The experiments was conducted using light and neutrally buoyant gases under three exhaust speed ratios. In this study, a scenario with a high-buoyancy gas with one exhaust speed ratio was selected to validate the CFD performance in this study. The detail of experiment setting can be found in the previous work [16], and brief introduction is given here.

The target building was considered as 20 m high in full scale. The target exhaust gas temperature was considered over 500 °C. The meteorological conditions at full scale were considered as the annual values in Tokyo, Japan. The ambient temperature was 15 °C and the wind velocity at the building height was 3 m/s.

As Fig. 1 shows, a cubic building model at a 1:50 scale ($H = 400$ mm) was set in the turbulent boundary layer. The origin of all the measurements was set at the center of the low edge of the leeward wall. The exhaust stack was a cylindrical chimney behind the building with a height of $h_s = 0.125H$ and diameter $d_s = 0.015H$. The wind direction was perpendicular to the windward wall of the building. The exhaust

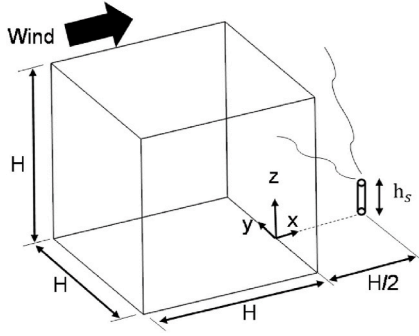


Fig. 1. Experiment setup.

velocity (W_s) was the same as the time-averaged streamwise velocity at the building's height (U_H).

The similarity between the model and full scales is important when conducting buoyant-gas dispersion experiments. The local Froude number ($Fr = \frac{\rho_s U_H^2}{(\rho_s - \rho_a) H g}$) [40] and the Reynolds number ($Re = U_H H / \nu$) [42] were used for the similarity of buoyancy gas dispersion and flow field, where g is acceleration due to gravity and the kinematic viscosity of air was $\nu = 1.47 \times 10^{-5} \text{ m}^2/\text{s}$ at 15°C . By setting $U_H = 0.68 \text{ m/s}$ and using the light gas ($\rho_s / \rho_a = 0.17$) in the wind tunnel experiment, Fr corresponded to the full-scale phenomenon with an exhaust gas temperature of 590°C . The relationship between density and temperature there was based on the standard atmospheric pressure. However, as U_H is restricted to a low value by Fr to reproduce a higher exhaust gas temperature in the full scale, Re was not strictly persevered between wind tunnel experiment and the full scale phenomenon. Meanwhile, previous researches [43] showed that if Re is large enough, the flow field will be self-similar with that in a higher Re , which is called Reynolds independence. For the flow field around an isolated building, Castro and Robins [43] suggested that Re should over 4000 to achieve Reynolds independence. In addition, Snyder [44] recommended that $Re \geq 11000$ is conservative. Tominaga and Stathopoulos [38] compared the measured velocity for $Re = 5700$ with those for $Re = 57000$, and confirmed that the effect of Re was very small. Moreover, Yoshie et al. [25] measured the flow field around high-rise building in non-isothermal boundary layer with $Re = 17000$. In this study, as $Re = 2.27 \times 10^4$, the flow field in the wind tunnel experiment can be considered self-similar with the full-scale phenomenon.

However, for the flow field measurement, the magnitude of velocity in the building wake under the condition with $U_H = 0.68 \text{ m/s}$ was smaller than the measurable range of constant temperature anemometer. However, we thought that experimental velocity data would be useful for roughly grasping the prediction accuracy of the flow field by CFD analysis. Therefore, although Fr cannot be strictly maintained, only the flow field measurement was conducted under a higher velocity ($U_H = 2.48 \text{ m/s}$), and results non-dimensionalized by U_H were used for the validation of velocity fields. In addition, as the gas emission may affect the flow field near the exhaust port, we decided to release neutral gas with an exhaust velocity increased in proportion to U_H during the flow field measurement.

Fig. 2 shows the inflow profile of the time-averaged streamwise dimensionless velocity (U/U_H) and turbulence intensity ($I_U = \sqrt{u'^2}/U$) [42]. The solid line represents a fitting curve based on the power law, where the power-law exponent (α) was 0.22.

For the concentration field, volume concentrations (C_{raw}) were measured at the vertical ($y/H = 0$) and horizontal ($z/H = 0.25$) planes. The results are presented and discussed based on the dimensionless concentration ($C = C_{raw}/C_0$), where $C_0 = q/H^2 U_H$ is the reference concentration. Here, q is the pollutant exhaust rate. The value of C_0 is 3.77 ppm.

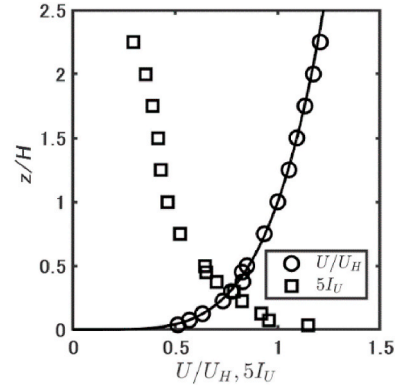


Fig. 2. Inflow time-averaged velocity and turbulence intensity.

3. CFD simulation setup

3.1. Computational domain and grid

Fig. 3 shows the simulation domain. The height and width of domain were the same as those in the wind tunnel. The distances from the inlet to the building and from the building to the outlet were set as five and ten times the building height, respectively, according to the Architectural Institute of Japan (AIJ) guidelines [45]. To reproduce the approaching flow profile in CFD simulation, the profiles measured in WTE (time-averaged velocity, turbulent kinetic energy, etc.) were prescribed at the inlet. Then, to reproduce the flow development between inlet and target bundling, we set the roughness (wood block cube with 30 mm) on the upwind side in the simulation domain with the same arrange in WTE. A polyhedral grid was used for its advantage in reproducing complex object shapes. The simulation domain was discretized into 8.4×10^6 grids based on the preliminary grid sensitivity check outlined in the [Supplementary data](#).

3.2. Numerical method and boundary conditions

The commercial CFD software STAR-CCM+ 12.02.010 was used. Table 1 lists the conditions of the numerical analysis and boundary conditions of the RANS and LES models.

Concerning the RANS model, a previous study [38] on the dispersion near an isolated building demonstrated that the realizable $k-\epsilon$ model [46] can more accurately predict both the time-averaged velocity and concentration compared with the standard $k-\epsilon$ [47], RNG $k-\epsilon$ [45], and shear stress transport (SST) $k-\omega$ models [49]. Therefore, the realizable $k-\epsilon$ 2-layer model was used to predict the dispersion of high buoyancy gas in this study. Residuals for all variables were below 10^{-4} .

For the LES, the wall-adapting local eddy-viscosity (WALE) model [50] was adopted because it could correctly reproduce the turbulence scaling near the wall without using a damping function. The WALE model adequately predicted the flow and concentration fields in the near field [51–53]. The time step was set to $\Delta t = 0.001 \text{ s}$ ($0.0017H/U_H$) as in the experiment. The computation was performed for 90 s ($153H/U_H$) before being time-averaged from the RANS result. The time-averaging lasted 90 s, which was the same as in the experiment. The sampling time was considered sufficient to obtain the time-averaged and fluctuation values. For the inlet boundary condition, the synthetic eddy method (SEM) [54] was used to generate synthetic turbulence. SEM considers turbulence as a superposition of eddies and represents the spatial and temporal characteristics of each eddy using specific shape functions. Specific first- and second-order one-point statistics and autocorrelation functions can be reproduced. In this study, the Reynolds stresses and turbulent length scale were approximated from the experiment. $\overline{u'v'}$ and $\overline{v'w'}$ were set to 0 as the theoretical values.

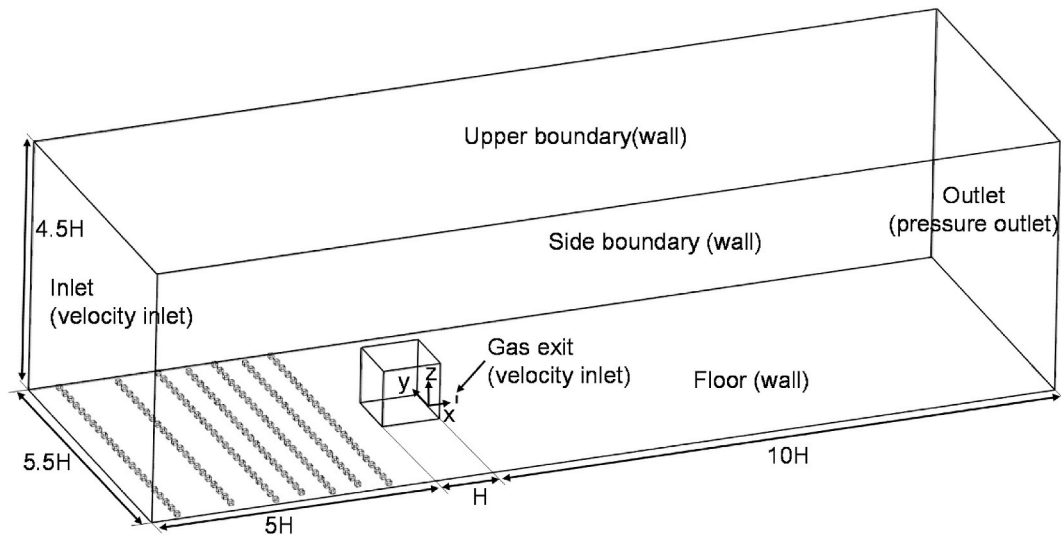


Fig. 3. Simulation domain.

Table 1
Numerical analysis and boundary conditions.

Simulation model	RANS	LES
Simulation domain	$16 H (x) \times 5.5 H (y) \times 4.5 H (z)$	
Simulation mesh	Polyhedral, 8.3×10^6	
Turbulence model	Realizable k- ϵ 2-layer model	Wall-adapting local eddy-viscosity model
Velocity pressure coupling/time marching	SIMPLE	PISO, second-order implicit scheme
Discretization scheme for advection term	Second-order upwind difference scheme	Bounded central differencing scheme: Central-differencing scheme0.15, Second-order upwind differencing scheme0.85
Discretization scheme for diffusion term	Second-order central differencing	
Inlet boundary condition	Time-averaged streamwise velocity, and turbulent kinetic energy from the experiment, turbulent dissipation rate according to AIJ guidelines [45]	Synthetic eddy method: Time-averaged streamwise velocity, Reynolds stresses, and turbulent length scale from the experiment
Exhaust-port boundary condition	Vertical velocity $W_s = U_H = 0.68 \text{ m/s}$, Mass concentration $\varphi = 1$, Gas temperature $T_s = 1766 \text{ K}$	
Outlet boundary condition	Zero gradient	
Other boundary conditions	Wall function, Adiabatic	

3.3. High-buoyancy gas modeling

In the gas modeling, the transportation equation of a passive scalar based on the mass concentration (φ) was solved. The value of φ at the exhaust port was set to 1. The CFD results were compared with the C measured in the WTE by transferring φ to volume concentration using $C = \frac{\rho_a \varphi}{\rho_s(1-\varphi) + \rho_a \varphi}$.

In the RANS model, the turbulent scalar flux ($\overline{u_i \varphi'}$) was based on the standard gradient-diffusion hypothesis (SGDH) as $\overline{u_i \varphi'} = -\frac{\nu_t}{Sc_t} \frac{\partial \varphi}{\partial x_i}$, where ν_t is the kinematic eddy viscosity. The value of Sc_t was set to 0.7 based on previous research on pollutant dispersion in the near field [55]. The effect of Sc_t on the high-buoyancy gas dispersion is presented in Section 4.5. The turbulent Prandtl number (Pr_t) was 0.9. In the LES, the

sub-grid scale Schmidt number (Sc_{SGS}) was set to 0.7.

For buoyancy modeling, instead of setting a passive scalar as a mass fraction combination of helium and ethylene, we set the temperature difference between the passive scalar and ambient air in CFD to reproduce the equivalent density difference between the light gas and ambient air in the WTE. This was because we aimed to examine the performance of this buoyancy modeling method for a future high-temperature gas dispersion simulation at full scale.

An incompressible model with the Boussinesq approximation is commonly used to relate the density difference and temperature difference. However, Zhou et al. [56] conducted a CFD simulation on the mixed convection of a high-temperature heat source in a building and observed that the Boussinesq approximation may lose applicability when the temperature difference is large. Therefore, a compressible model must be considered.

However, a fully compressible model requires high temporal and spatial resolutions to calculate the acoustic-wave propagation, which causes numerical stability problems and increases the computational burden. Meanwhile, since the Mach number can be considered to approach zero in urban flow, the energy in acoustic waves is significantly smaller than that of the fluid convection velocity and thermodynamic internal energy.

Therefore, an incompressible ideal gas model was considered a suitable method for this study. In this method, the ideal gas law is used to relate the temperature and density. Since the pressure is constantly set as atmospheric pressure, the density (ρ) is only affected by temperature (T) and can be calculated using $\rho T = \rho_a T_a$, where T_a is the ambient temperature. In previous studies, the incompressible ideal gas model provided reasonable predictions on high-temperature, high-buoyancy gas and thermal stratification [56–58].

In this study, T_a was set to 300 K. The gas temperature (T_s) at the exhaust port was set to 1766 K to reproduce equivalent buoyancy ($\rho_s / \rho_a = 0.17$) in the WTE.

4. Results

4.1. Flow field

Fig. 4a–c shows the distributions of the time-averaged streamwise velocity (U/U_H) in the vertical plane ($y/H = 0$). As explained in Section 2, the experimental velocity data was measured in a high U_H with neutral gas emission. For the simplicity, the results in the horizontal plane ($z/H = 0.25$) can be found in Supplementary data. The LES provided a close result with the experiment. Meanwhile, RANS exhibited a

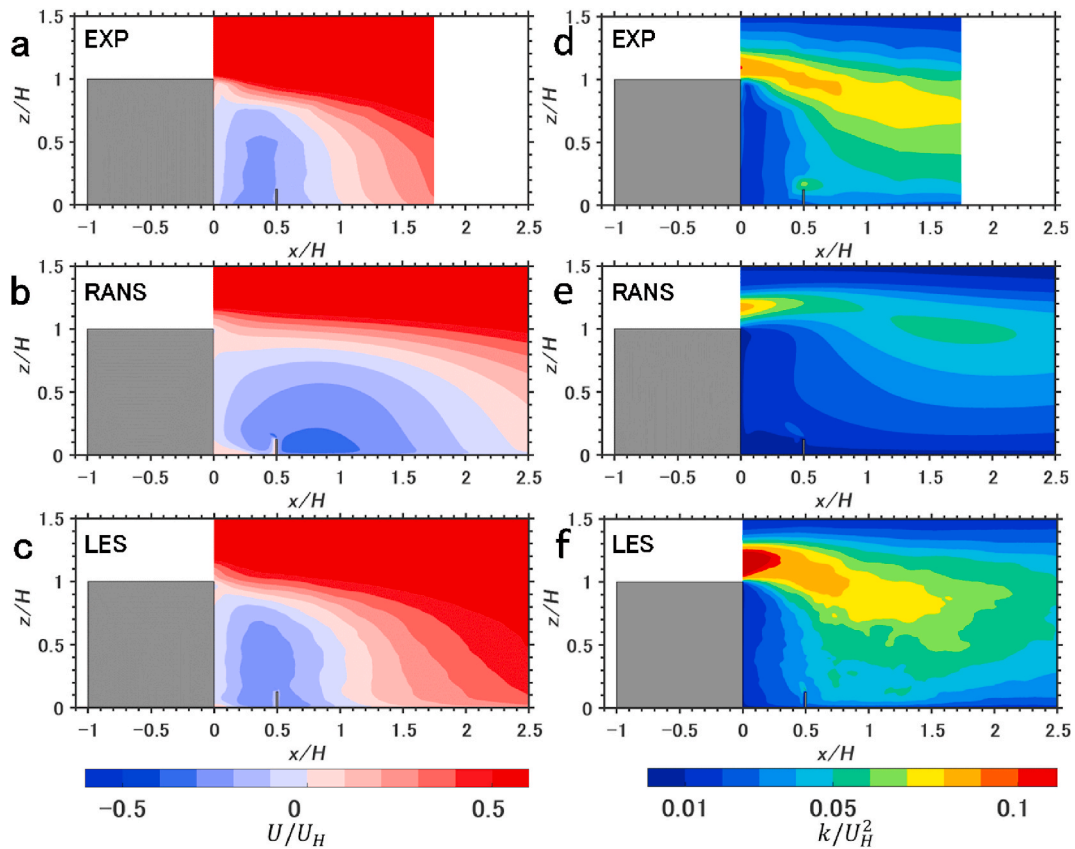


Fig. 4. Distribution of time-averaged streamwise velocity (U/U_H) and turbulent kinetic energy (k/U_H^2) in the vertical plane ($y/H = 0$) obtained using the EXP, RANS model and LES.

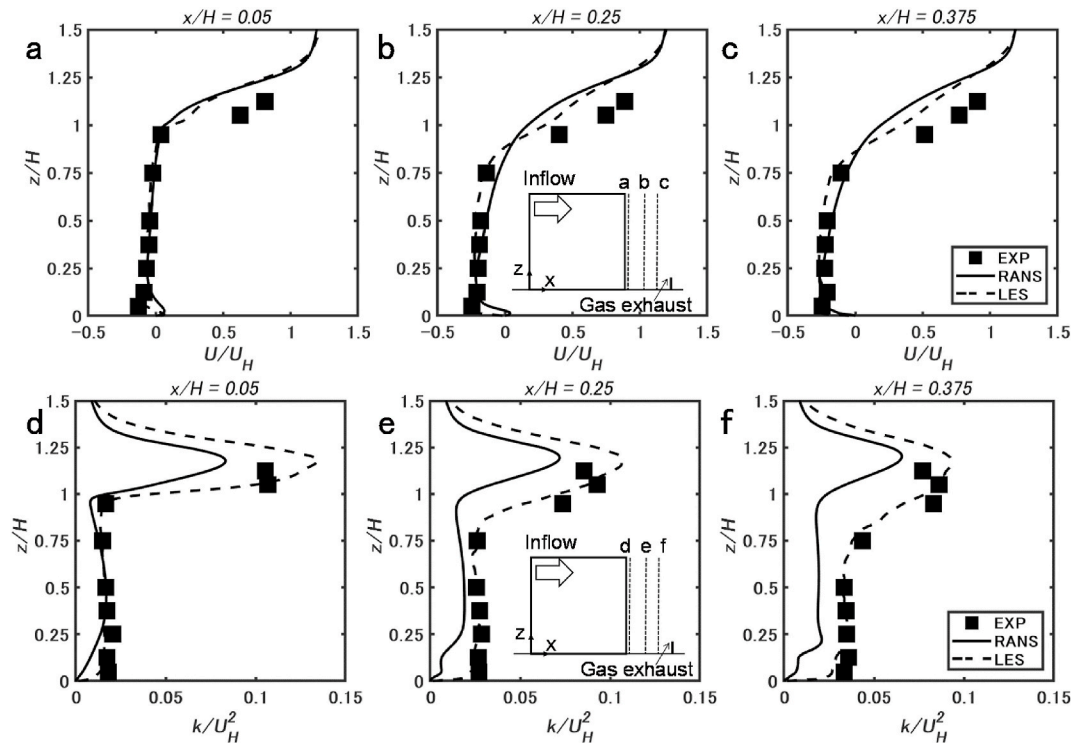


Fig. 5. Vertical profiles of time-averaged streamwise velocity U/U_H and turbulent kinetic energy k/U_H^2 in the vertical plane ($y/H = 0$). The experimental values were measured under a higher reference velocity ($U_H = 2.48 \text{ m/s}$) with the emission of neutral gas.

longer reattachment length and a larger recirculation region. Fig. 5a–c shows the vertical profiles of U/U_H in the vertical plane. The comparison with the experimental values indicated that the RANS model and the LES adequately reproduced the reverse flow in the building's wake. However, the RANS model and the LES underestimated the separation flow at the building's height. The place has a strong shear with high fluctuation and is hard for both measurement and prediction, and reason of discrepancy between the experiment and CFD models needs to be further investigated.

Fig. 4d–f shows the distributions of the turbulent kinetic energy (k/U_H^2) in the vertical plane ($y/H = 0$). In the experiment, a large k/U_H^2 was observed in the separation region and near the exhaust port. The LES exhibited a larger value than the RANS model in both the separation and cavity regions. This was because the LES could reproduce the periodic large-scale fluctuation behind the building [17]. When focusing on the vertical profiles of k/U_H^2 (Fig. 5d–f), we observed that the LES accurately predicted the experimental values. The RANS model predicted an agreeable k/U_H^2 tendency with the experiment, while the value was generally underestimated, particularly near the ground ($z/H < 0.25$). Although not shown here, the results in the horizontal plane ($z/H = 0.25$) had a similar tendency as the results above. The LES reproduced the separation flow at the building edge ($y/H = 0.5$) more successfully than the RANS model (Fig. S1 and Fig. S2 in Appendix B). The tendency of the LES to exhibit a smaller velocity and larger turbulent kinetic energy than the k- ϵ model agreed with previous research on an isolated building [21].

4.2. Time-averaged concentration

Fig. 6 shows the comparison of the distribution of the time-averaged concentration ($\bar{C}$) in the vertical plane. In the experiment, the pollutant was blown towards the building's leeward wall after emission owing to wake flow, and the pollutant was conveyed windward at the roof height. Therefore, a larger value was observed between the exhaust port and the building's leeward wall.

Note that Sc_t was set to 0.7 in the RANS model. The large concentration region was accurately reproduced. However, the RANS model tended to overestimate $\bar{C}$ in the building's wake, as indicated by Tominaga and Stathopoulos [21]. In the LES, $\bar{C}$ had smaller values than the RANS model, which were comparable to the experimental data. In particular, the plume in the LES rose higher and the ground-level $\bar{C}$ was much lower than in the RANS model. Meanwhile, the LES had a shorter distribution in the x-direction at roof height.

Fig. 7 shows the vertical profiles of $\bar{C}$ in the vertical plane. In the wind tunnel experiment, the measurement was conducted several times. Therefore, the time-averaged concentration and concentration standard deviation can be little different each time. The markers denote the mean value of all measured results, and the error bars denote the range (from minimum to maximum) of measured values. The same denotation will be used in the following figures of profiles. In general, the LES had a closer prediction compared with the RANS model on the high-buoyancy gas dispersion. In the region near the exhaust port ($x/H = 0.375$), both the RANS model and LES exhibited good agreement with the experimental values. The LES had a smaller $\bar{C}$ than the RANS model at the peak location in the profile. At the center between the exhaust port and the building's leeward wall ($x/H = 0.25$), the RANS-predicted $\bar{C}$ variation along the z-direction corresponded with the experimental data, while the value was overestimated at $z/H < 0.3$. The LES exhibited a better agreement than the RANS model, although its $\bar{C}$ was slightly underestimated at $z/H \geq 0.5$. Near the building's leeward wall ($x/H = 0.05$), the LES indicated a decrease in $\bar{C}$ when approaching the ground, while this was not apparent in the RANS model. Moreover, the difference in peak location between the RANS and LES methods increased with the distance from the exhaust port.

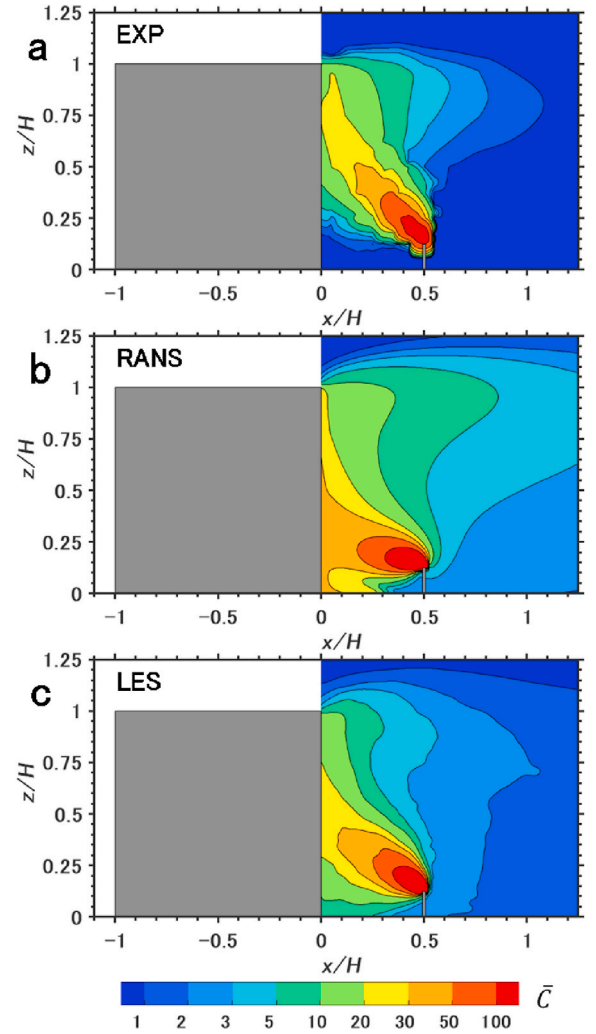


Fig. 6. Distribution of time-averaged dimensionless concentration $\bar{C}$ in the vertical plane ($y/H = 0$).

4.3. Concentration fluctuation

Since exhaust gas is considered to have high temperatures and high concentrations of pollutants, its instantaneous values are important as well as time-averaged values. Fig. 8 shows the distribution of the concentration-standard-deviation (C_{SD}) in the vertical plane. In the experiment, the distribution of C_{SD} was similar to that of $\bar{C}$, indicating a higher-value region near the exhaust port. The values of C_{SD} were generally larger than those of $\bar{C}$, indicating that the concentration fluctuation intensity ($C_{SD}/\bar{C}$) was larger than uniform. The LES adequately predicted the distribution of C_{SD} , particularly for the high-concentration region. At roof height, the LES exhibited a slightly compact distribution compared with the experiment. The vertical profiles (Fig. 9) indicated that the LES successfully predicted both C_{SD} values and the peak locations in the profiles.

4.4. Effect of the turbulent Schmidt number in the RANS model

Turbulent Schmidt number Sc_t can have a direct effect on the scalar dispersion when using the standard gradient-diffusion hypothesis (SGDH) on turbulent scalar flux modeling in RANS computations. One of the main reasons for the RANS model overestimating the time-averaged concentration, as indicated in Section 4.2, was that the periodic large-scale fluctuation behind the building could not be reproduced, leading to the underestimation of turbulent diffusion [38]. As a result, a smaller

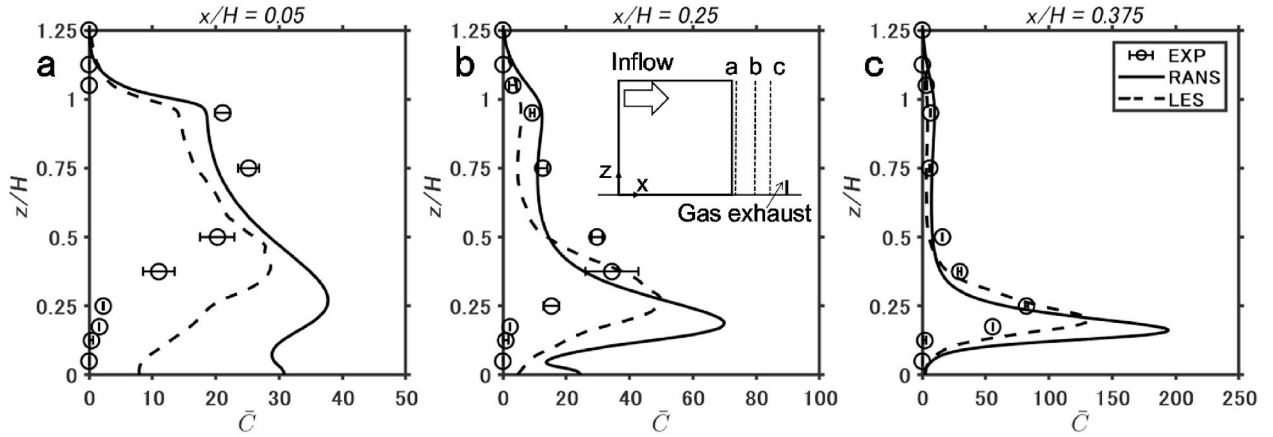


Fig. 7. Vertical profiles of time-averaged dimensionless concentration $\bar{C}$ in the vertical plane ($y/H = 0$). The markers denote the averaged values, and the error bars denote the ranges of measured values during the measurements.

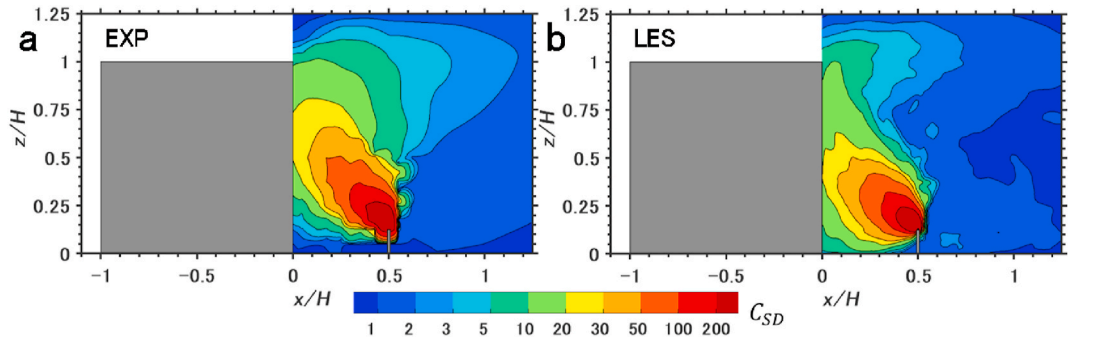


Fig. 8. Distribution of dimensionless concentration-standard-deviation C_{SD} in the vertical plane ($y/H = 0$).

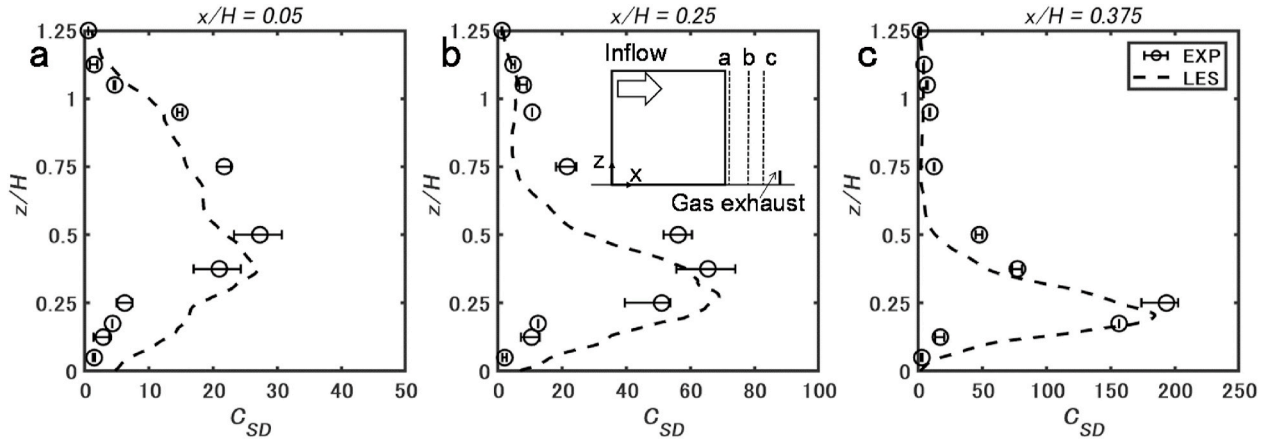


Fig. 9. Vertical profiles of dimensionless concentration-standard-deviation C_{SD} in the vertical plane ($y/H = 0$). The denotations can be referred to Fig. 7.

value of Sc_t was suggested to promote turbulent diffusion, thereby alleviating the overestimation of $\bar{C}$ [55].

For non-buoyant gas, the value of Sc_t was commonly selected from 0.3 to 1.0 for dispersion around buildings [24,36]. However, only a few studies focused on the effect of Sc_t on high-buoyancy gas, particularly on the local dispersion characteristics. Since the buoyancy force depends on the local concentration of high-buoyancy gas, an investigation is required to provide a suitable selection of Sc_t values to predict high-buoyancy gas. For the calculation in this study, we selected Sc_t from 0.3 to 1.5.

Fig. 10a–c depict the vertical profiles of $\bar{C}$ in the vertical plane for

different values of Sc_t . In general, smaller values of $\bar{C}$ were observed when Sc_t was small, which was consistent with non-buoyant gas. However, when Sc_t was large, the ground-level $\bar{C}$ decreased and the peak location in the profiles increased, which was comparable to the experimental values. This was because large values of Sc_t lead to less turbulent diffusion; therefore, the gas accumulated near the exhaust port, resulting in a more effective buoyancy to transport the gas upward. In addition, at $x/H = 0.375$, the difference between scenarios with various values of Sc_t were only observed near the peak value. However, the $\bar{C}$ profiles were completely different at $x/H = 0.05$, indicating that the Sc_t effect on the dispersion in the cavity is intensified with the distance from

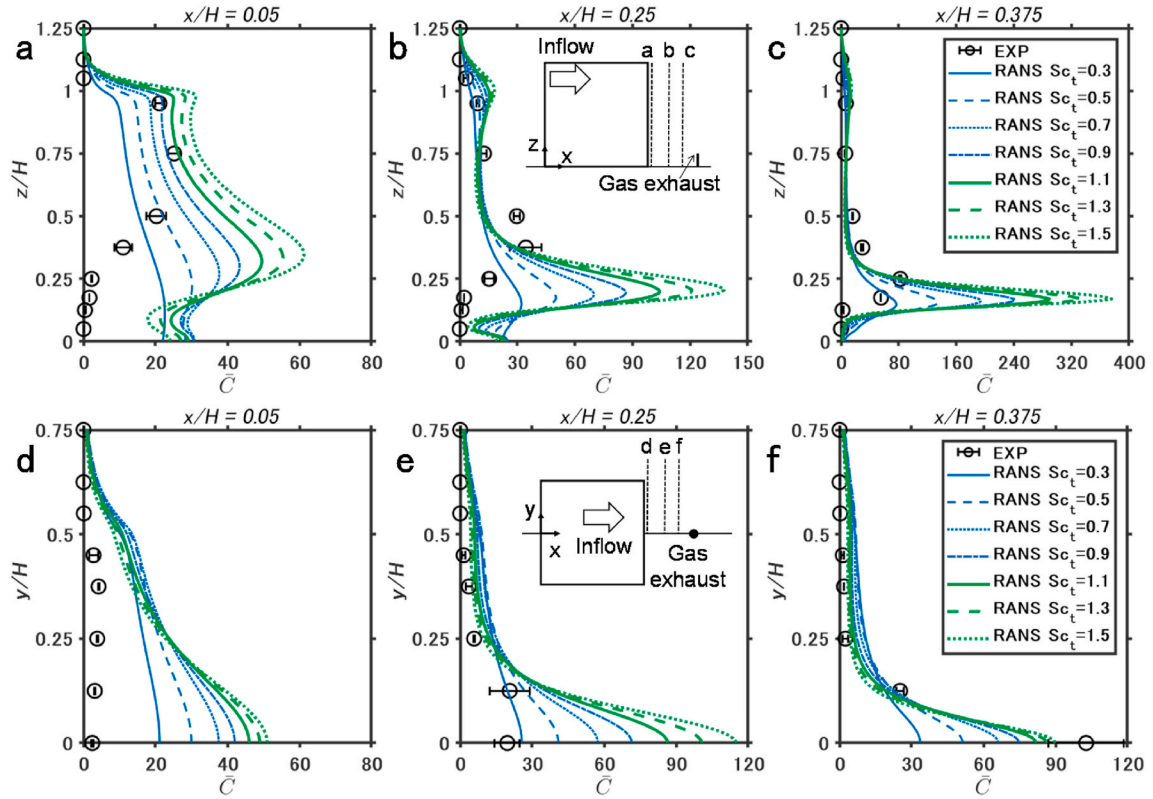


Fig. 10. Vertical (spanwise) profiles of time-averaged dimensionless concentration $\bar{C}$ in the vertical plane ($y/H = 0$) (and horizontal plane $z/H = 0.25$) based on different values of turbulent Schmidt number Sc_t .

exhaust port.

In the horizontal plane (Fig. 10d–f), scenarios with large values of Sc_t exhibited higher values of $\bar{C}$ at the building centerline and lower values of $\bar{C}$ at further locations in the spanwise direction compared with cases with lower values of Sc_t . A large Sc_t exhibited better agreement with the experimental data near the exhaust port, while the opposite was observed near the building leeward wall. This suggested that Sc_t is dependent on the location, and that producing agreeable results in all places with a constant Sc_t is difficult. In addition, although it seemed that a smaller Sc_t could provide results comparable to those of the experimental values, the limitation was that the concentration profiles were not consistent with experimental data compared with the LES.

5. Discussion

Validation metrics were used to quantify the CFD prediction accuracy results with the experimental values. The following metrics were used in this study to discuss the performance on the concentration field: the fraction of the prediction within a factor of 2 of the experimental values (FAC2), fractional bias (FB), and normalized mean square error (NMSE) [59]. The metrics are defined as follows:

$$FAC2 = \frac{1}{N} \sum_{i=1}^N n_i, \quad n_i = \begin{cases} 1, & 0.5 \leq \frac{CFD_i}{EXP_i} \leq 2 \\ 1, & EXP_i \leq W \text{ and } CFD_i \leq W \\ 0, & \text{else} \end{cases} \quad (1)$$

$$FB = \frac{\overline{EXP} - \overline{CFD}}{0.5(\overline{EXP} + \overline{CFD})} \quad (2)$$

$$NMSE = \frac{\overline{(EXP_i - CFD_i)^2}}{\overline{EXP} \times \overline{CFD}} \quad (3)$$

where EXP_i and CFD_i are the experimental and CFD prediction values of the dimensionless concentration for sample i , respectively, and N is the number of data points. The overbar represents the mean value of the entire dataset. For FAC2, the allowed absolute difference W was set to 0.05. The ideal values are 1 for FAC2 and 0 for FB and NMSE. Previous research suggested that $FAC2 > 0.5$, $|FB| < 0.3$, and $NMSE < 4$ can be accepted for concentration prediction [60]. In this study, 57 and 15 measurement points on the vertical plane ($y/H = 0$) horizontal plane ($z/H = 0.25$) were considered for metrics. To focus on the performance of the region where exhaust gas may pose the largest threat, we selected the validation points between the exhaust port ($x/H = 0.5$) and the building's leeward wall ($x/H = 0$).

Table 2 shows the validation results for $\bar{C}$ for both the RANS and LES models and C_{SD} for the LES. The results are presented individually for the vertical and horizontal planes. In general, the results in the vertical plane were more acceptable than those in the horizontal plane, indicating the difficulty in predicting horizontal dispersion. The comparison of the performance of the LES and RANS models on $\bar{C}$ indicated that the LES exhibited better results than the RANS scenarios and had all validation metrics results in the acceptance range except for FAC2 and FB in the horizontal plane. In addition, the LES exhibited good results for C_{SD} .

For the RANS results in the vertical plane, better performance for FAC2 and FB was observed when Sc_t was small. However, a negative FB and lower FAC2 were observed at $Sc_t = 0.3$ than at $Sc_t = 0.5$, implying an excessive underestimation on $\bar{C}$. Although NMSE was less sensitive for Sc_t than with FAC2 and FB, it exhibited a better result at $Sc_t = 1.1$ because the overestimation and underestimation cancelled each other out. Meanwhile, in the horizontal plane, the best performance for FAC2 was observed at $Sc_t = 1.5$. This was because a large Sc_t led to a small $\bar{C}$ at the locations away from the centerline ($y/H > 0.25$), resulting in prediction values closer to the experimental values. Meanwhile, better results of FB and NMSE were observed at $Sc_t = 0.3$ as the overall $\bar{C}$ was

Table 2

Validation metrics for time-averaged dimensionless concentration and dimensionless standard deviation of the concentration. The underlined values indicate the best performance in the RANS computations.

Validation metrics	Vertical plane ($y/H = 0$)			Horizontal plane ($z/H = 0.25$)		
	FAC2	FB	NMSE	FAC2	FB	NMSE
Ideal value	1	0	0	1	0	0
Acceptance range	> 0.5	$ FB < 0.3$	< 4	> 0.5	$ FB < 0.3$	< 4
LES ($\bar{C}$)	0.58	-0.08	0.88	0.20	-0.84	1.13
LES (C_{SD})	0.60	0.21	0.83	0.80	-0.15	0.16
RANS ($\bar{C}$) $Sc_t = 0.3$,	0.58	<u>0.10</u>	2.58	0.13	<u>-0.78</u>	<u>1.06</u>
RANS ($\bar{C}$) $Sc_t = 0.5$,	<u>0.63</u>	-0.18	2.51	0.13	-0.91	1.48
RANS ($\bar{C}$) $Sc_t = 0.7$,	0.52	-0.32	2.53	0.13	-0.96	1.88
RANS ($\bar{C}$) $Sc_t = 0.9$,	0.49	-0.38	2.43	0.13	-0.94	2.12
RANS ($\bar{C}$) $Sc_t = 1.1$,	0.46	-0.42	<u>2.41</u>	0.27	-0.94	2.41
RANS ($\bar{C}$) $Sc_t = 1.3$,	0.42	-0.45	2.45	0.33	-0.92	2.67
RANS ($\bar{C}$) $Sc_t = 1.5$,	0.35	-0.48	2.53	<u>0.47</u>	-0.91	2.91

smaller. In general, a small Sc_t increased the overall prediction accuracy in both the vertical and horizontal planes.

6. Conclusion

To evaluate the CFD prediction of high-temperature exhaust gas from emergency generators, we conducted steady RANS and LES experiments on a high-buoyancy gas dispersion around an isolated cubic building. The accuracy of the flow and concentration fields were validated using a wind tunnel experiment.

In the wind tunnel experiment, light gas was used to reproduce high buoyancy in the high-temperature exhaust gas. The density ratio of light gas and ambient air in the wind tunnel experiment was interpreted as 590 °C at full scale based on the local Froude number. To reproduce the equivalent density difference in the CFD, we set the temperature difference between the passive scalar and ambient temperature. An incompressible ideal gas model was adopted to demonstrate the relationship between the density and temperature differences because it provides a higher accuracy than the incompressible model with the Boussinesq approximation when the temperature difference is large. The following conclusions were drawn from the results of this study:

- (1) The RANS model exhibited good agreement with the experimental value of the time-averaged velocity, but underestimated the turbulent kinetic energy. For the concentration field, RANS predicted the region of high time-averaged concentration near the exhaust port but overestimated ground-level values ($z/H < 0.25$). In addition, the validation metrics showed that RANS with $Sc_t = 0.7$ can provide acceptable FAC2 (0.52), FB (-0.32) and NMSE (2.53) in the vertical plane ($y/H = 0$). Meanwhile, RANS showed difficulty in predicting dispersion in the horizontal plane ($z/H = 0.25$).
- (2) The LES accurately predicted both time-averaged velocity and turbulent kinetic energy. For the mean concentration field, the LES exhibited good results in the vertical plane (FAC = 0.58, FB = -0.08, NMSE = 0.88). However, the LES overestimated the mean concentration in the horizontal plane. On the other hand, the LES gave encouraging results on the concentration standard deviation

in both vertical (FAC = 0.60, FB = 0.21, NMSE = 0.83) and horizontal plane (FAC = 0.80, FB = -0.15, NMSE = 0.16).

- (3) Investigations on the effect of the turbulent Schmidt number (Sc_t) in the RANS model indicated that a small Sc_t increased the overall prediction accuracy in the vertical plane (i. e., FAC = 0.63, FB = -0.18, NMSE = 2.51 at $Sc_t = 0.5$). However, a large Sc_t compensated for the overestimation of ground-level time-averaged concentration, and provided a better FAC2 (0.47 at $Sc_t = 1.5$) in the horizontal plane. Nevertheless, the concentration profiles were not consistent with experimental data compared with the LES.

In this study, we focused on the evaluation of the RANS and LES methods in predicting time-averaged concentration near the exhaust port. Meanwhile, the effect of buoyancy may have not fully been considered, and future research will focus on buoyancy modeling in the transport equations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.buildenv.2020.107478>.

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